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Effect of noble metal ions dopants on solar photoelectrochemical water splitting and electrochemical supercapacitive performance of $BiVO_4$ hollow tubes

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ABSTRACT

Owing to their poor electrochemical properties, photoelectrochemical (PEC) solar water splitting and electrochemical energy storage activity of $BiVO_4$ are presently limited in their applications. One of the effective methods for improving electrochemical activity is the doping of noble-metal ions into the host lattice. The doping of noble-metal ions can improve absorption in visible light and support the direct transfer of energy via hot electrons. In this study, the synthesis of Au-doped $BiVO_4$ nanostructured catalysts with different concentrations of Au ions was prepared via a template-free ultrasonication technique, and it's both solar-driven PEC water splitting and electrochemical storage properties were examined. The optimal doped photoelectrode exhibited a lower resistance to charge transfer than other photoelectrodes, including a significant enhancement in photocurrent density (0.102 $mAcm^{-2}$), which is approximately 4.1 times superior over undoped photoanode. Furthermore, the optimal doped electrode exhibited an electrochemical capacitance that is 2 times greater than that of the pure electrode. The excellent electrochemical behavior and PEC properties of the optimal doped electrode were attributed to the increase in surface area and suitable band gap. Consequently, enables the dispersion of electron at the interface of electrode/electrolyte. Solar PEC solar water splitting and supercapacitor performance of the optimal doped electrodes were significantly improved by the proper incorporation of Au ions.

1. Introduction

Electrochemical energy storage and conversion systems with ultra-inclusive performances provide the technologies required to address overwhelming global energy challenges [1]. In recent years, solar energy, which is the most abundant energy source, has been extensively studied for electrical and thermal power generation [2]. Hydrogen was proposed as one of the most favorable fuel choices for the storage and distribution of energy obtained from renewable sources. Photoelectrochemical (PEC) water splitting has been established as a more suitable option for energy storage (H_2) than electricity, and solar energies have been deliberated as one of the most encouraging technological ideas, owing to their prospective superior efficacy and eco-friendly features [3]. Furthermore, because PEC solar water splitting, as a zero emission route, is currently a crucial topic [4], the development of a photocatalyst that can harness the entire electromagnetic spectrum is required to enhance the proficiency of solar water splitting [5]. Therefore, numerous photocatalysts have been established and evaluated for water oxidation over the years [6]. Among them, $BiVO_4$ is one of the most active photocatalysts, which is harmless and a relatively abundant material [7]. Owing to its relatively low band gap, which allows the use of additional effective visible light, suitable conduction, and valence band positions, $BiVO_4$ enhances the kinetics of water oxidation. However, owing to its reduced charge transport, slow

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water oxidation kinetics, and high recombination charge carrier rate, $BiVO_4$ is significantly limited in its performance [8]. Therefore, to improve performance of these electrodes, numerous methods have been adopted, such as morphology control [9], noble-metal doping [10], construction of heterojunctions [11], and decoration of water oxidation catalysts [12].

Among them, the incorporation of noble metal ions into nanostructured metal oxide catalysts can efficiently enhance its water splitting performance. The sensitization of plasmonic materials can significantly improve PEC competencies owing to the photogenerated charge carrier separation that overwhelms their recombination rate [13]. Moreover, owing to the local surface plasmon resonance (LSPR) phenomena, Au noble-metal insertion would prolong the absorption of visible light ability. Hence, the excitation of LSPR will assist the PEC activity led by the local photon energy transfer close to the semiconductor [14,15]. Besides the LSPR effect, for the effective oxidation of water to oxygen, the Au metal can set up the $BiVO_4$ photogenerated electrons and leave the same amount of holes. Furthermore, Au^+ ions could act as trapping sites for holes that decrease the recombination rate of electron-hole pairs and encourage the transfer of interfacial charge to improve PEC efficiency [16]. Owing to its sustainable influence of the LSPR near the visible light region and strong trapping electron capability, Au doping has garnered substantial attention [6,17].

A significant feature of electric fields is that they are spatially nonhomogenous and exhibit the greatest field strength influence on the nanostructures. Hence, this implies that the SP-induced formation of the charge carrier pairs should be supreme in the semiconductor, which is contiguous to Au. Therefore, from this perspective, for the first time, we report the synthesis of highly efficient Au-doped $BiVO_4$-nanostructured photoanodes with different dopant concentrations of Au ions using an ultrasonication method. Furthermore, we investigated their structural, morphological, optical, and electronic properties, together with their PEC solar water oxidation and electrochemical energy storage properties.

2. Experimental section

2.1. Pure and Au-doped $BiVO_4$ synthesis procedure

Nitric acid (5 mol/L) was mixed in 20 mL of DI water (solution A), while Bi $(NO_3)_3 \cdot 5H_2O$ (9 g) was dissolved in solution A. NaOH (6.5 mol/L) was mixed in 20 mL of DI water (solution B) and NH_4VO_3 (2.1 g) was dissolved in solution B. Then, under ultrasonication, the entire solution B was gently mixed in solution A. NaOH was used to regulate the pH level at 6. Different Au concentrations (1 mL, 3 mL, and 5 mL) of $HAuCl_4$ were separately added dropwise to the solution obtained from the mixture of solutions A and B. Subsequently, the achieved yellow precipitate was repeatedly washed with DI water and ethanol, and then desiccated at 70 °C. Finally, the acquired samples were heated in air for 2 h at 500 °C. Hereinafter, the synthesized Au-doped samples are referred to as 1-BV (1 mL), 3-BV (3 mL), and 5-BV (5 mL).

2.2. Characterizations

An X-ray diffractometer (X'Pert PRO, PANalytical) was used to examine the crystal phases of the prepared samples. A transmission electron microscope (TEM; H-7600, HITACHI), field emission scanning transmission electron microscope (G^2 F20 S-Twin, Tecnai), and field emission scanning electron microscope (FE-SEM; S-4800, Hitachi) were employed to examine the morphology of the samples. The X-ray photoelectron spectra (XPS) was recorded using a high-performance XPS system (Thermo Scientific MultiLab 2000). An imaging spectrometer (iHR 550, Horiba), at an excitation wavelength of 325 nm, was used to examine the photoluminescence spectra. Brunauer–Emmett–Teller (BET) surface area analysis and Barrett–Joyner–Halenda (BJH) pore-size distribution analysis were employed to examine the surface area. The electrochemical properties were examined using a standard three-electrode potentiostat (Sp-300, Biologic) in a KOH (0.1 M) electrolyte. E(RHE) was calculated in the equation E(RHE) = E(Ag/AgCl) + 0.1976 V + 0.0591 pH, where E(RHE) and E(Ag/AgCl) represent the potentials (E) as a function of a reversible hydrogen electrode (RHE) and an Ag/AgCl electrode, respectively.

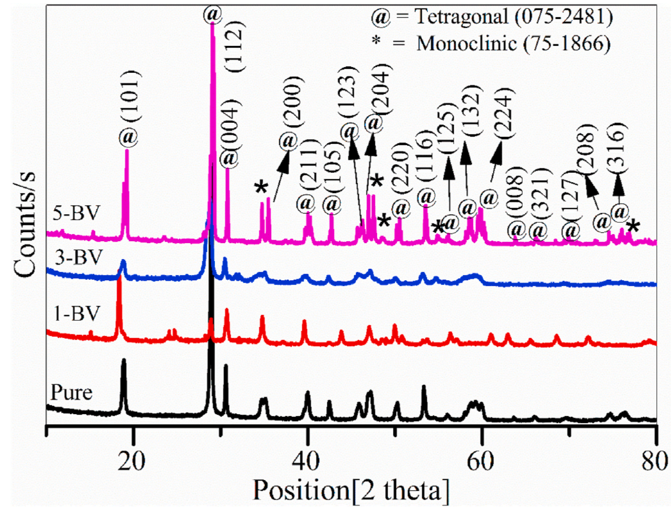

Fig. 1. X-ray diffraction patterns of pristine and Au-doped $BiVO_4$ nanostructures.

3. Results and discussion

3.1. X-ray diffraction analysis

The crystal structure of the prepared samples was determined using X-ray diffraction (XRD). Fig. 1 presents the XRD patterns of the pure and Au-doped (with different Au contents) $BiVO_4$ nanostructures. From the XRD patterns, it can be clearly observed that all samples exhibit distinctive tetragonal phase diffraction peaks of the $BiVO_4$ (JCPDS no: 75–2481). Furthermore, no contamination-associated peaks were observed in the doped samples, which clearly indicates that the dopant ions were successfully incorporated into the host lattice. As can be observed in Fig. 1, as the dopant content increases, a lower diffraction angle shift is observed in the case of 1-BV and 3-BV samples, and a further increase in the dopant concentration (5-BV sample) triggers a slight increase in the diffraction angle shift, thus representing an increase in the c-axis lattice parameter. The identified peak shift is due to the dissimilarity in the ionic radius of the host lattice and dopant ion, which indicates that dopant ions are doped interstitially in the host lattice [18]. Furthermore, in the higher-concentration doped sample (5-BV) case, monoclinic phase-related peaks were identified at 34.7°, 47.5°, 48.6°, and 76.9°, respectively. Similar results have been reported in the literature [19,20].

Scherrer's equation was used to calculate the average crystallite sizes of the prepared samples: $D = \frac{0.9\lambda}{\beta\, Cos\theta}$, where λ, β, and θ represent the X-ray wavelength, full width at half maximum (FWHM) diffraction peak, and diffraction angle, respectively. The calculated average crystallite sizes for pure, 1-BV, 3-BV and 5-BV samples were 65.5, 73.5, 79.0, and 81.2 nm, respectively. The average crystallite size increased from 65.5 nm to 81.2 nm when the dopant content was increased. A similar result was reported by Sahu et al. [21]. Such an increase in crystallite size with increasing doping concentrations could be attributed to the increase in internal structural defects, which enhances the rate of growth, as well as the movement of atoms at grain boundaries [22].

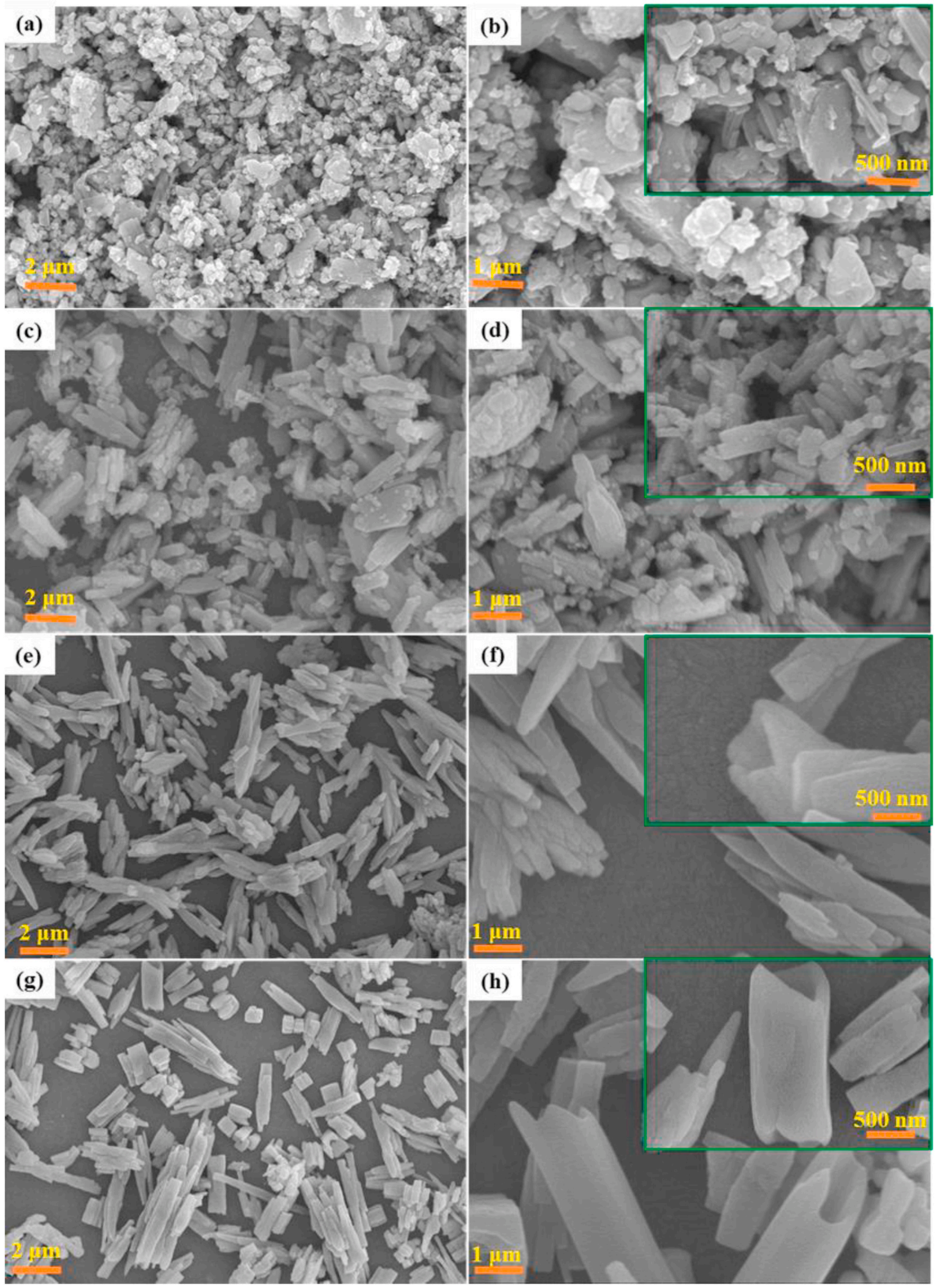

Fig. 2. SEM images of (a, b) Pure, (c, d) 1-BV, (e, f) 3-BV and (g, h) 5-BV samples with different magnifications.

3.2. Surface morphology analysis

A FE-SEM was used to study the surface analysis of the pure and doped $BiVO_4$ samples, and the obtained SEM micrographs are presented in Fig. 2 with different magnifications. Fig. 2(a) and (b) show the pure $BiVO_4$ SEM images with flake-like surface morphologies. However, in the 1-BV-sample case, its morphology was altered from flacks to mixtures of both flacks and hollow shaped tubes, as illustrated in Fig. 2(c) and (d). In the case of the 3-BV and 5-BV samples, as the concentration increased, the morphology completely changed from flacks to hollow tube-like morphologies, as shown in Fig. 2(e)–(h). The alterations in the morphologies of the doped samples could be attributed to numerous issues, such as the shape of the intrinsic structure of the crystal, precursor content, temperature, and technique [23,24]. Catalyst decomposition initiates similar nucleation by developing an impetuous combination enhancement to reduce the surface energy, thus creating hollow tube-shaped structures anisotropically [25]. The morphological modifications with the dopant could possibly alter the binding energies of ligand−metal, as well as the surface energies of precise crystallographic facets, which leads to the development of anisotropically larger particles (into different structures) than those obtained in Ostwald's ripening period [26,27]. Furthermore, EDS technique is used in order to examine the investigation of elementals present in the prepared samples along with the elemental mapping and the corresponding data is shown in Fig. 3. As a result, EDS data analysis indicate the presence of Bi, V, O and Au as a major elements. Fig. 4 illustrates the TEM and high-resolution transmission electron microscopy images of pure and optimal doped (3-BV) $BiVO_4$ nanostructures. Flacks and tube-like

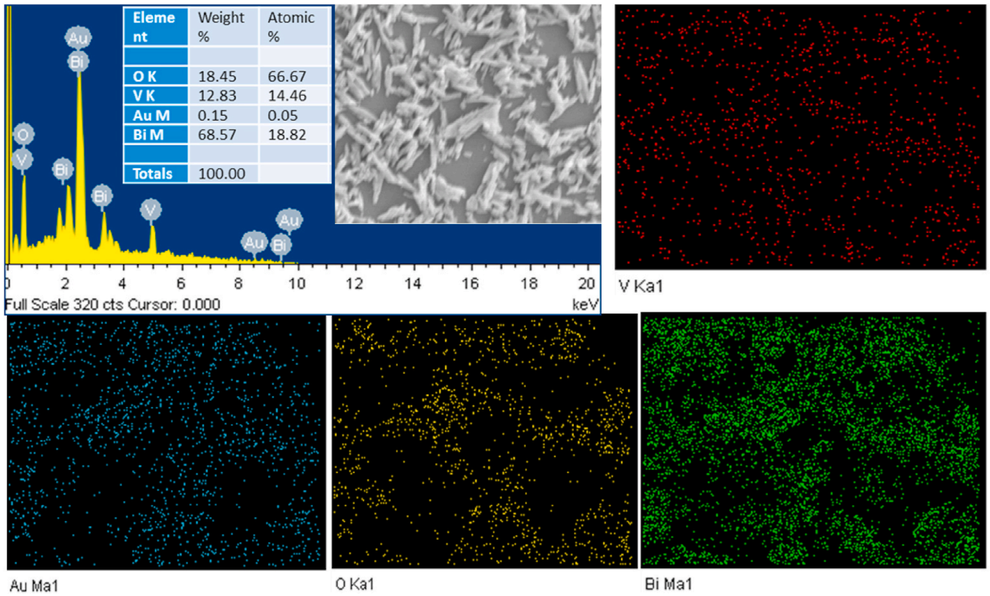

Fig. 3. EDS spectra and elemental mapping of 3-BV sample.

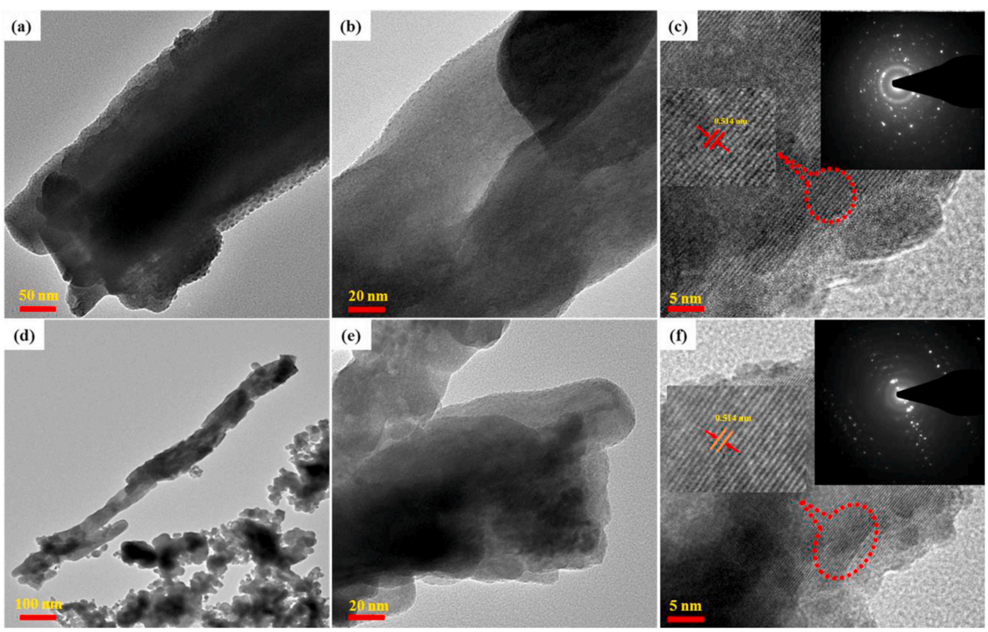

Fig. 4. TEM and HR-TEM images of (a, b) Pure and (c, d) 3-BV sample (Inset: Magnified HR-TEM image and SAED pattern).

morphologies, supported by the SEM analysis, are observed from the TEM images. Clear lattice fringes with d-spacing of 0.514 nm, which correspond to the (112) lattice planes of tetragonal $BiVO_4$, are observed in both samples. Moreover, the selected area electron diffraction pattern (shown in the inset figure) could be indexed to the tetragonal planes of $BiVO_4$, thus indicating high crystallinity.

3.3. BET analysis

Generally, the improved specific surface area increases the number of active sites and reduces the photo-excited charge carrier's distance to a reaction site, which eventually leads to a greater number of electrons and holes available for the reaction. Hence, nitrogen adsorption-desorption measurements were performed to estimate the textural properties and inner pore architectures of the prepared materials, and the results are presented in Fig. 5. As can be observed from the BET curves, the prepared undoped and Au-doped $BiVO_4$ nanostructures exhibited the characteristic type-IV isotherm for all samples and showed hysteresis loops at the P/P0 ranges of 0–1.0, which is distinctive of the mesoporous material. Similar observations have been reported in the literature [28–30] The BET and BJH models were used to evaluate the surface area, pore volume, and distribution of pore size, with their values presented in Table 1. Consequently, the 3-BV sample exhibited a better surface area, pore volume, and pore size than the other samples. Hence, the optimal doped sample might exhibit better solar water splitting abilities than the other samples.

3.4. Optical absorption studies

Fig. 6(a) illustrates the UV–vis diffuse reflectance spectra (DRS) of

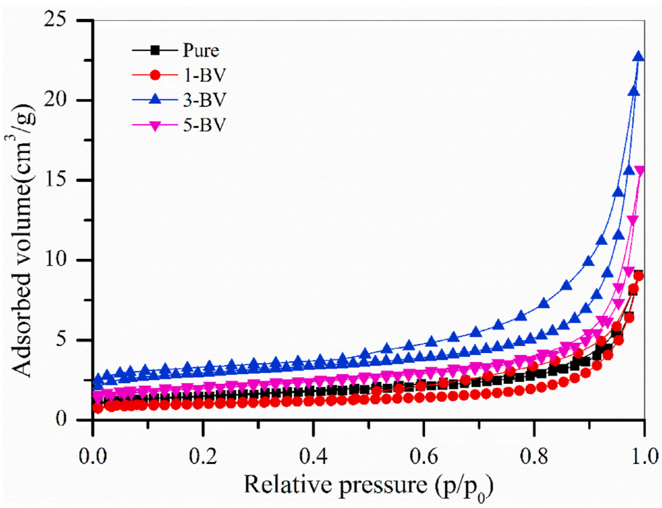

Fig. 5. N_2 adsorption/desorption isotherms of pristine and Au-doped $BiVO_4$ nanostructures.

Table 1
Surface area (S_{BET}), pore size, and pore volume of prepared undoped and Au-doped $BiVO_4$ nanostructures.

Sample name	S_{BET} m^2/g	Pore Size (nm)	Pore Volume (cm^3/g)
Pure	3.56	14.7	0.01321
1-BV	5.141	15.1	0.01583
3-BV	10.65	22.2	0.0335
5-BV	6.89	16.9	0.02412

the pure and Au-doped $BiVO_4$ nanostructures. It was established that the pure $BiVO_4$ exhibits an absorption band edge at approximately 533 nm in the visible region with a bandgap of 2.34 eV [31]. Furthermore, in case of the Au-doped $BiVO_4$ nanostructures, the absorption band edges of the prepared nanostructures moved towards longer wavelengths after doping with Au nanoparticles, which significantly enhanced the visible-light harvesting ability. The absorption band edges of 1-BV, 3-BV, and 5-BV were determined to be approximately 590, 600, and 580 nm, respectively. Subsequently, the band gaps estimated from the absorption edges are 2.11, 2.08, and 2.15 eV, respectively. Furthermore, Au-doped samples exhibit a broad peak between 640 and 660 nm, which is may be attributed to the agglomeration of gold particles and corresponding localized surface plasmon resonance (LSPR) absorption has been observed. Similar studies have been reported in the literature [32–35], which significantly confirms the establishment of a strong interface between the plasmonic Au and $BiVO_4$ lattice [32]. Berlin et al. [36], observed a surface plasmon resonance (SPR) peak at 610 nm in their Au doped ZrO_2, and when the doping amount of Au increased to 12%, the SPR peak shift towards a higher wavelength (654 nm) and broadens. The broadening and shift of SPR peak are due to the change in host environments of the metal oxides. Kreibig et al. [37], reported that the broadening of SPR peaks is due to the damping, which is determined by the distance traveled by electron scattering, which in turn depends on the size of noble metal particles. Furthermore, both blue and red shifts plasmon has been observed in several studies [38,39], In addition, the bandgap energy of the pure and Au-doped $BiVO_4$ nanostructures was measured via Tauc plot analysis and presented in Fig. 6(b). From Fig. 6 (b), a declining bandgap was observed with an increase in dopant concentration in the case of 1-BV and 3-BV samples. Further increased the dopant content, increased band gap was observed in 5-BV sample, which could be attributed to the interface between the Au metal and $BiVO_4$ [40]. Furthermore, the measured bandgap energy values for pure, 1-BV, 3-BV, and 5-BV were determined to be 2.37, 2.09, 2.02, and 2.14 eV, respectively. Therefore, the shift in the bandgap confirms the doping of Au into $BiVO_4$ at substitutional sites.

3.5. X-ray photoelectron spectra analysis

XPS analysis was carried out to examine the elemental compositions and chemical states of the prepared Au-doped $BiVO_4$ nanostructures. The XPS of the pure and Au-doped $BiVO_4$ nanostructure (3-BV) are presented in Fig. 7. The survey spectra (Fig. 7(a)) and high-resolution XPS of Bi 4f, V 2p, O 1s, and Au 4f (Fig. 7(b)–(e)) further confirm that the surface of the prepared nanostructure consists of Bi, V, O, and Au elements. The high-resolution Bi 4f XPS (Fig. 7(b)) of pure $BiVO_4$ and 3-BV sample exhibited two prominent peaks at 164.18 and 158.86, as well as 164.12 and 158.79 eV, respectively, which corresponds to the Bi $4f_{7/2}$ and Bi $4f_{5/2}$ spin states, and reveals the existence of Bi^{3+} [41]. The peaks associated with V $2p_{3/2}$ and V $2p_{1/2}$ are observed at 516.52 and 524.18 eV, respectively, for pure $BiVO_4$, as well as 516.47 and 524.15 eV, respectively, for 3-BV, as shown in Fig. 7(c). Moreover, these peaks are the characteristic peaks of the V^{4+} oxidation state [42]. The high-resolution spectra of O 1s (Fig. 7(d)) possess binding energies of 529.64 and 529.58 eV for pure $BiVO_4$ and 3-BV, respectively, which correspond to the characteristic peak of lattice oxygen [43]. The high-resolution XPS illustrated in Fig. 7(e) presents the binding energy of Au 4f. The peak positions centered at 83.4 and 87.3 eV were determined and ascribed to Au $4f_{7/2}$ and Au $4f_{5/2}$, respectively [44]. The Au XPS analysis suggests that the no peak of metallic Au is observed, which suggests that Au exist in the univalence state and substitute for host lattice [45,46]. A slight negative shift was observed owing to the robust electronic interface between the Au and the host lattice [47]. Furthermore, the binding energies of Bi 4f, V 2p, and O 1s for the Au-doped $BiVO_4$ nanostructure are smaller than those for pure $BiVO_4$, which is strong evidence for the doping of metal Au into $BiVO_4$ at substitutional sites.

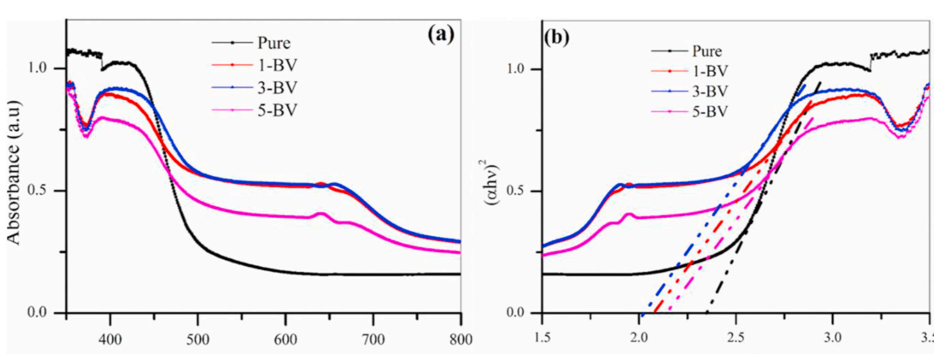

Fig. 6. (a) Optical absorption spectra, and (b) optical band gaps of pristine and Au-doped $BiVO_4$ nanostructures.

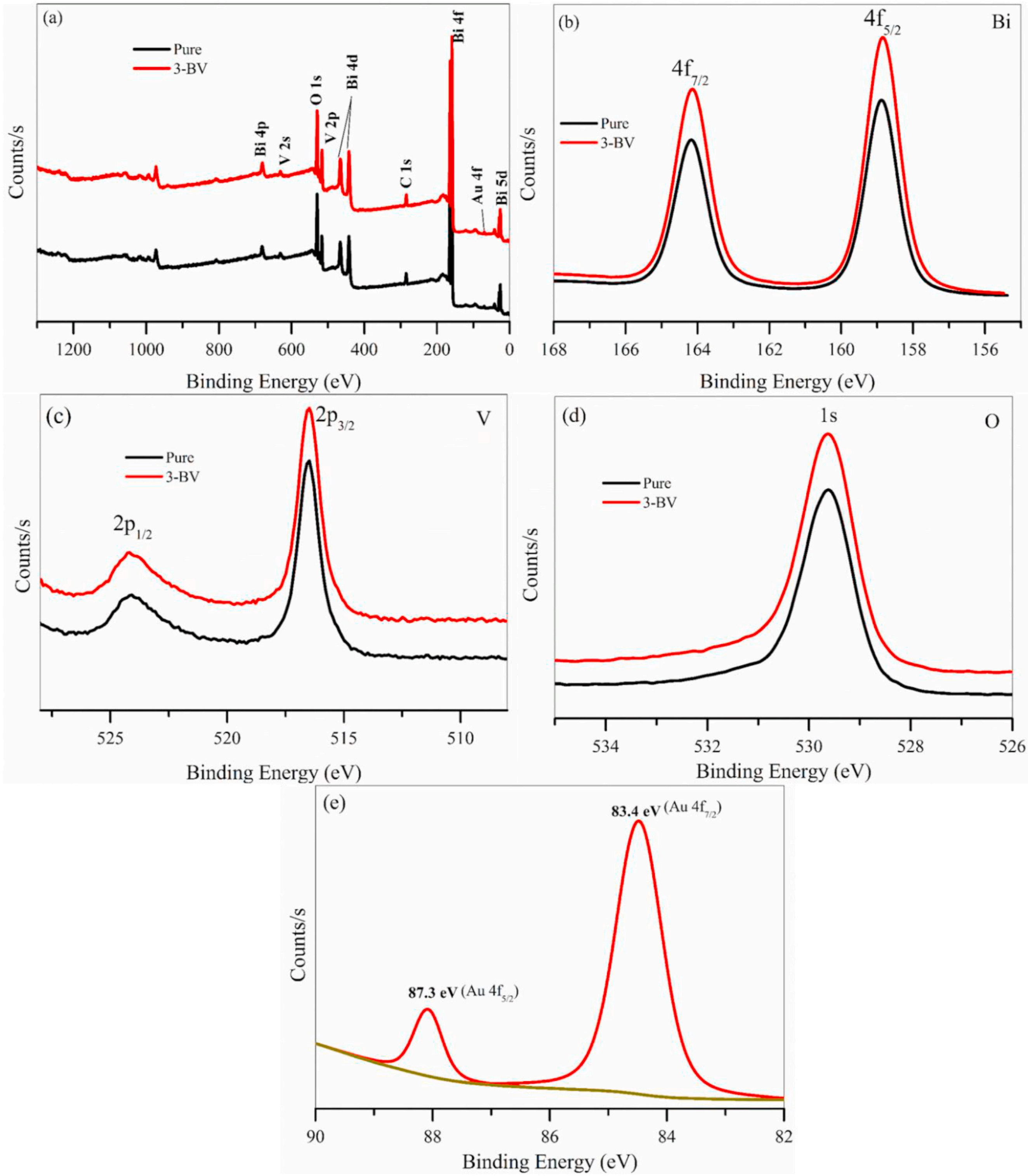

Fig. 7. XPS spectra of (a) survey, (b) Bi, (C) V, (d) O, and (e) Au of pure and 3-BV samples.

3.6. Photoelectrochemical analysis

Electrochemical impedance spectroscopy (EIS) was adopted to examine the characteristics of the interfacial or bulk region-bound charges in the semiconductors. Fig. 8(a) illustrates the Nyquist plots of all the prepared photoelectrodes under visible light irradiation. Generally, the Nyquist plot arch indicates the charge-transfer kinetics of the working electrode, as the semicircle diameter reveals the resistance of the charge transmission. As can be observed from Fig. 8(a), the Nyquist semicircle diameter decreased more significantly after the Au doping. A smaller arc radius indicates that the most active charge transfer will occur within the semiconductor. Hence, it was observed that the Au-doped electrodes exhibited a weaker resistance to the reaction and active transfer of charge under light illumination than the pure $BiVO_4$ electrode. Among the doped electrodes, the 3-BV electrode exhibited a reduced charge resistance than the 1-BV and 5-BV electrodes. The Au

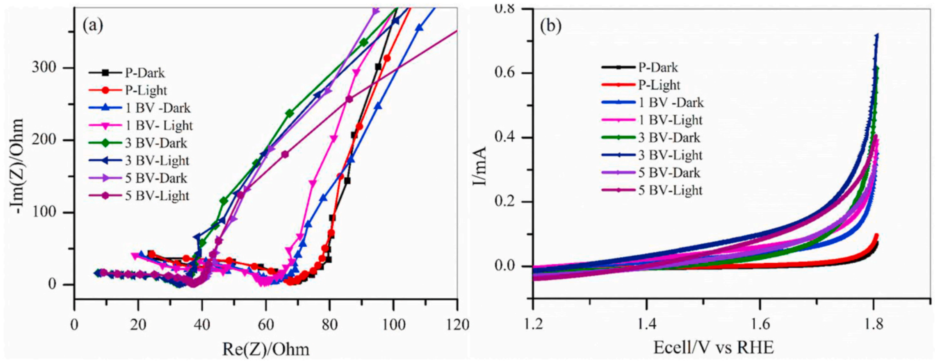

Fig. 8. a) EIS spectra, and (b) LSV response of pristine and doped electrodes under dark and light illumination.

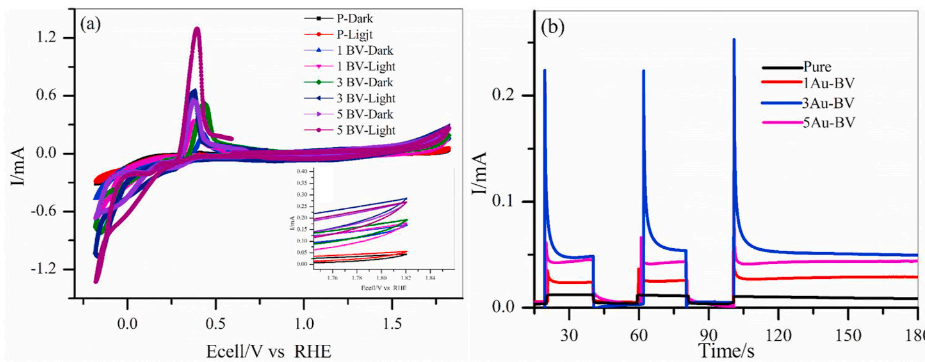

Fig. 9. (a) CV curves, and (b) transient photocurrent response of pure and doped photoanodes.

doping significantly reduced the charge resistance from 67 to 32 Ω, and the overall charge recombination within the host lattice was significantly reduced after the incorporation of the Au ions to the host lattice. Similar results have been reported in the literature [48,49]. Consequently, the charge transfer efficacy and separation of the photoexcited charge carriers were significantly enhanced under light illumination in the Au-doped electrode than in the undoped electrode. In summary, Au doping reduced the recombination rate of charge carriers and successively advances the generation of the photocurrent and effective separation of charge carriers, which results in the enhancement of PEC solar water splitting.

The photo induced I–V curves were assessed using linear sweep voltammograms (LSVs) to examine the PEC solar water splitting performance of the prepared photoanodes, and the results under visible light and without irradiation are presented in Fig. 8(b). As can be observed in Fig. 8(b), the undoped photoanode exhibited a current of 1.34 V (vs. RHE) under light illumination. However, with the increase in bias voltage under both light and dark conditions, the photocurrent densities of all doped photoanodes increased more significantly than those of the undoped photoanode. The photocurrent density was imitated at 0.92 V (vs. RHE) under light irradiation in the case of the optimal doped photoanode. Furthermore, the difference in photocurrent density increased from 0.024 $mAcm^{-2}$ to 0.102 $mAcm^{-2}$. A photocurrent density of 102 $mAcm^{-2}$, which is approximately 4.1 times greater than that of the pristine photoanode, was obtained for the optimal doped photoanode. Among all doped photoanodes, the optimal doped photoanode (3-BV) exhibited a significantly enhanced photocurrent density owing to the improved visible light region absorption and

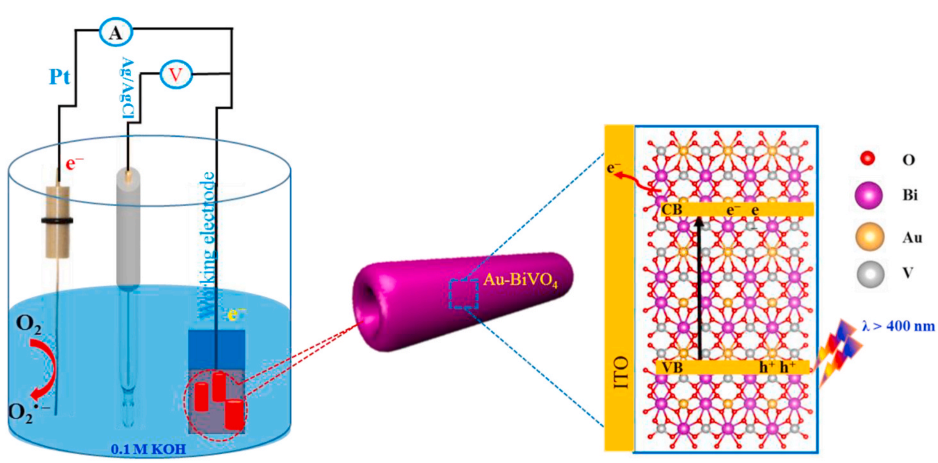

Fig. 10. Solar water splitting schematic representation of Au-doped $BiVO_4$ nanostructures.

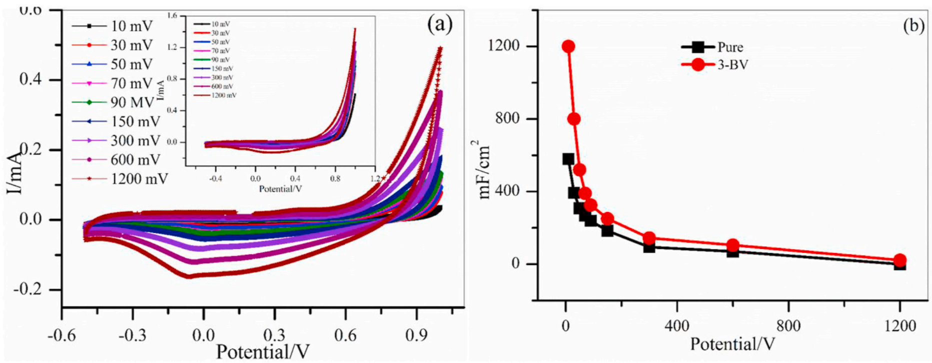

Fig. 11. CV curves, and (b) capacitance values of pristine and 3-BV electrodes at different scan rates.

photogenerated separation of charge carriers, which resulted in a more substantial enhancement of its photocurrent density than that of the undoped photoanode. This confirmed that the doping of Au into the host lattice helps to enhance PEC activity, which results in better light absorption and surface passivation. Consequently, this leads to a reduction in the recombination of the charge carriers [50,51].

Similarly, cyclic voltammetry (CV) analysis was used to examine the photocurrents of the prepared photoelectrodes under dark and light illuminations, and the results are presented in Fig. 9(a). From the curves, it is observed that the 3-BV electrode exhibited a more enhanced photocurrent density than other electrodes, and also a superior current density (0.284 $mAcm^{-2}$) to that of the pure electrode (0.057 $mAcm^{-2}$) under light irradiation, as it was determined to be approximately 4.9 times superior to the undoped electrode. The attained photocurrent density of the 3-BV electrode could be attributed to the abridged resistance of charge transfer, which minimized the recombination rate of charge carriers and enriched the hole generation and movement of electrons. Additionally, anodic peaks were identified in all samples between 0.39 and 0.44 V, and the oxidation peak is attributed to the oxidation of $Bi^0 \rightarrow Bi^+$.

The photocurrent response of the prepared photoelectrodes to different concentrations of Au is illustrated in Fig. 9(b). From Fig. 9(b), it can be observed that the photocurrent progressively increased with an increase in the dopant content of the Au-doped photoelectrodes than that of the undoped photoelectrode. Among the doped photoelectrodes, photocurrent density is very poor in the case of the 1-BV and 5-BV electrodes, and the response is delayed. However, in the case of the 3-BV electrode, the delayed response and photocurrent density improved significantly. Furthermore, the photocurrent density increased from 0.01 $mAcm^{-2}$ to 0.22 $mAcm^{-2}$, which is approximately 22 times greater than the photocurrent density of the rest. Because Au ions are incorporated into the host lattice, additional electrons will be transferred in the doped electrodes than in the undoped electrode. Under an external electric field, electrons will be transmitted to the opposite electrode, thus increasing the density of the photocurrent.

Fig. 10 illustrates the possible mechanism for the PEC activity using the Au-BV photoanode. It is obvious that the Au doping into the host lattice increases the concentration of donors and provides a good carrier transport passage [52]. Instantaneously, the significantly reduced recombination rate of charge carriers owing to the variation of the interfacial energy levels enhances the separation of the charge and interfacial transfer, and thus inhibits the recombination of charge carriers [53]. Therefore, the photoexcited electrons can rapidly and strongly separate and migrate to the fluorine-doped tin oxide (FTO) conductive substrates. This permits the electrons gathered in the FTO conductive substrates to move through an external circuit to the counter electrode (Pt electrode) and contribute in the hydrogen evolution response. Regarding these remarkable circumstances, the collected holes can effectively oxidize the water into oxygen at the working photoanode. Therefore, owing to the Au doping, the Au-doped $BiVO_4$ photoanode exhibited outstanding PEC activity.

3.7. Supercapacitor analysis

The electrochemical properties of the prepared electrodes were examined in a standard three-electrode potentiostat for CV and EIS experiments as an electrode material for supercapacitors. A typical three-electrode cell consists of working electrodes (prepared electrodes), Ag/AgCl (reference electrode), and Pt (counter electrode), in a 0.1 M KOH aqueous solution at room temperature. Fig. 11(a) presents the results of the as-prepared undoped and optimal doped (3-BV) electrodes at various scan rates (10–1200 mV). From the curves, it can be observed that all scan rates of the CV curves are analogous, thus indicating that the electrodes exhibited a reversible nature. The CV-curve profile indicates that the mechanism (charge-discharge) of the electrode via a Faradaic reaction (metal ions), together with the electrolyte, enhanced the reaction rate. Similarly, the area of the CV curves indicates the total gathering of charge via the Faradaic and non-Faradic responses. The Faradaic reaction indicates the movement of ions with a surface-bound redox capacitance, whereas the non-Faradaic reaction indicates the effect of the dual-layer capacitance. Fig. 11 (a) demonstrates the CV consequences of the undoped and optimal doped electrodes. In general, owing to the outstanding nature of the capacitive electrodes, the CV curves exhibited symmetrical capacitive behaviors with a pseudo-rectangular shape [54]. In this study, the CV effects of the optimal doped electrode exhibited a rectangular shape and relatively increased its current level compared to undoped electrode.

Therefore, the optimal doped electrode exhibits total capacitance owing to its quasi-capacitance behavior, which is associated with the reversible reaction in the electrolyte. The CV curves are recorded at different scan rates, and peak (anodic and cathodic) intensities increase with the current density and sweep rate, as shown in Fig. 11(a). Moreover, peak locations remain unchanged, thus indicating that ions are transferred in both directions. Therefore, at the electrode-electrolyte interfaces, the redox reactions are sustained in the electrolyte. Owing to the hydroxyl functional groups present in it, the material exhibits a pseudo-rectangular behavior with a certain deviation [55]. Therefore, the CV analysis of the doped electrode may exhibit better electrochemical properties.

The specific capacitance of the prepared electrodes was calculated using the following equation [56]: $C = \frac{I \int dV}{2^{*}V^{*}m^{*}\Delta V}$, where C $I \int dV$, m, V, and ΔV represent the specific capacitance, CV-curve area, active material mass, scan rate, and potential window, respectively.

The calculated specific capacitance values of the undoped and optimal doped electrodes at different scan rates are presented in Fig. 11 (b). At a scan rate of 10 mVs^{-1}, the value of the calculated specific capacitance was increased from 578.64 $mFcm^{-2}$ to 1200.14 $mFcm^{-2}$ for undoped and optimal doped electrodes. These values evidently show that the optimal doped electrode is approximately 2 times better than

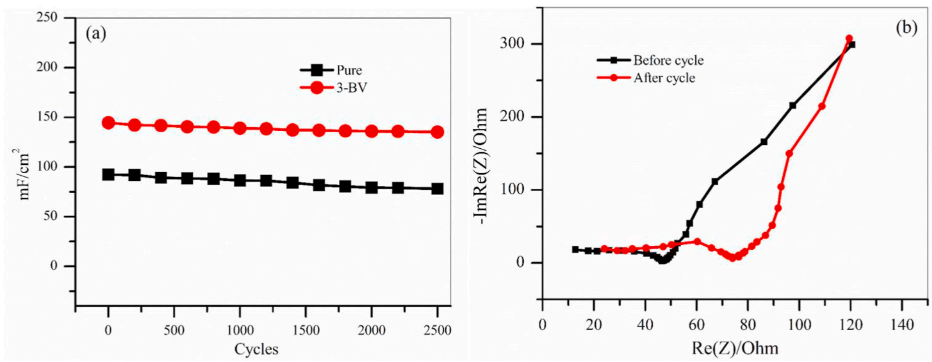

Fig. 12. (a) 2500 cycles stability of pristine and 3-BV samples, and (b) Nyquist plots obtained from the 3-BV electrode before and after the long-term cycling measurements.

that of the pure electrode. The improved performance of the doped electrode is evident in the fact that the doped electrode possesses extra active sites and exhibits a pseudo-capacitive nature and double layer. Additionally, the electrical conductivity of the doped electrode is enhanced owing to reduced internal resistance. Cycling activity is one of the most important features of a supercapacitor device. Therefore, to study the stability, 2500 cycles were recorded at a fixed scan rate (300 mVs^{-1}), and the results are presented in Fig. 12(a). From Fig. 12(a), it is evident that the cycles are significantly influenced by capacitance. Initially, a decrease in specific capacitance was observed in both electrodes, then stable capacitance values of approximately 140 and 94 $mFcm^{-2}$ appeared for the 3-BV and pure electrodes, respectively. The optimal doped electrode exhibited larger specific capacitance values even in cycles, which was approximately 1.7 times better than that of the pure electrode. Moreover, the electrode exhibits an outstanding cycling stability with a capacity retention of 90% after 2500 cycles. The above analysis indicates that the doped electrode exhibits exceptional electrochemical super capacitive properties, which could be attributed to the doping of the noble metals and nanostructures of the host material.

Fig. 12(b) illustrates the EIS of the optimal doped electrode before and after 2500 cycles. From Fig. 12(b), it can be observed that the EIS shows that the semicircle at high frequency R_{ct} is related to the charge transmission procedure at the electrode–electrolyte interface, and the straight line with a slope Z_w is associated with controlling the diffusion procedure. As can be observed from Fig. 12(b), although the EIS exhibits similar ohmic resistances before and after cycling, the semicircle after cycling is much larger than that before cycling. A similar observation was reported by Wu et al. [57]. Moreover, no substantial changes were noticed in the impedance curve after continuous cycling, thus indicating the outstanding stability of the electrode. The results verify the high consistency and stability of the optimal doped electrode.

4. Conclusions

In conclusion, Au-doped $BiVO_4$ nanostructures were successfully synthesized using the ultrasonication technique and studied for their PEC water oxidation properties as electrochemical supercapacitors. The optimal doped electrode exhibited an improvement in visible light absorption owing to the doping of Au ions. In addition, the optimal doped electrode exhibited a substantial improvement in the PEC performance via speed transfer and the separation of photoexcited charge carriers at the interface, and attained an enhanced photocurrent density. Under visible light irradiation, a photocurrent density of 102 $mAcm^{-2}$, which is approximately 4.1 times greater than that of the undoped photoanode, was obtained for the optimal doped photoanode. The specific capacitance of the 3-BV electrodes (1200.14 $mFcm^{-2}$) exhibited more significant enhancement than the undoped electrode (578.64 $mFcm^{-2}$), which was approximately 2 times greater than that of the pure electrode at a scan rate of 10 mVs^{-1}. The long-term cycling consistency of the 3-BV electrode was examined by CV analysis under ambient conditions. The electrode exhibited outstanding cycling stability with a capacity retention of 92% after 2500 cycles, thus verifying the outstanding cycling permanence of the electrode. We believe that Au-ion doping can be further optimized to improve the absorption of light and PEC activity. This successful approach could be useful for water splitting and supercapacitors using noble-metal doping for a variety of electrode materials.

CRediT authorship contribution statement

Ch. Venkata Reddy: Investigation, Methodology, Writing – original draft. **I. Neelakanta Reddy:** Conceptualization, Experimental, Visualization. **K. Ravindranadh:** Methodology. **Bhargav Akkinepally:** Planning, Methodology. **Fernando Alonso-Marroquin:** Writing – review & editing. **Kakarla Raghava Reddy:** Writing – review & editing, Supervision. **Bai Cheolho:** Writing – review & editing, Supervision. **Jaesool Shim:** Funding acquisition, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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